The results — a completed four-setting study

The empirical realisation of the [research programme]() in [ad hoc semantic reconciliation](). Four operational settings, built, run against a validated gold standard, and reported, with a master report drawing them into one synthesis.

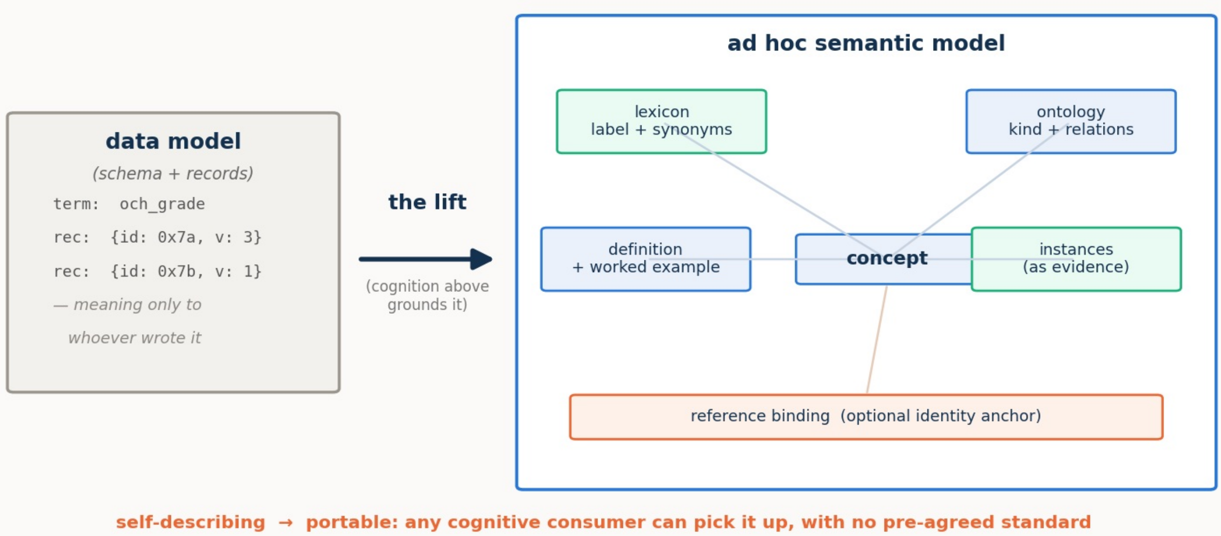
> *Part of the larger project — the theory and programme that motivate this study are in `../proposal/`; the interactive demonstrations are in `../demos/`.*

The central finding, established across all four settings: **it is cognition that completes a reconciliation.** Descriptor methods — matching names, then names plus a gloss — carry it to a ceiling and stop, historically leaving the rest to a standard or a person. Where both systems can reason, the remainder is closed **autonomously**, with no model agreed in advance and no human in the loop, and only as cognition recedes does a residual have to be referred onward. A thin, published **reference** can substitute for cognition or supply an inert side the information it lacks — but it reaches information and stops at authority. And the **pragmatic** layer — what a reconciled thing is *for*, whether it matters, who decides — is the frontier the descriptor methods never reach: decisive for meaning, and itself bounded by the agent's capability.

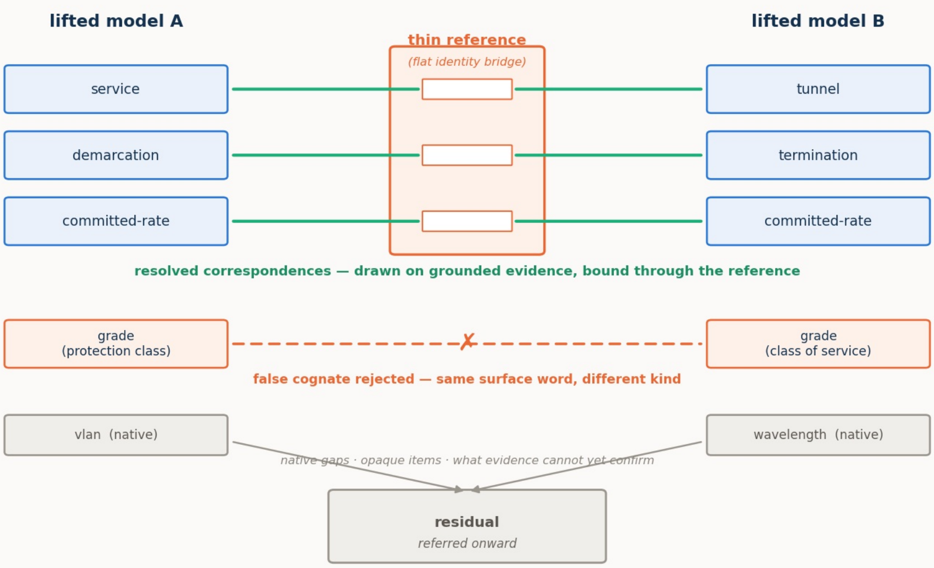
Start here — the master report

reports/MASTER_REPORT.md · **PDF** — the synthesis that sits atop the four settings: the idea (the lift, portable semantic models, the cognition spectrum, the thin reference), each setting distilled to its essentials, and the findings gathered so they can be read as one result, with two tables mapping every finding to the process stage it acts on.

The lift — a data model, meaningful only to its author, becomes an ad hoc semantic model that carries its own meaning



Reconciliation over two lifted models — grounded correspondences bound through a thin reference, a rejected cognate, and the residual referred onward



The four settings

Each setting takes the same frame to a new operation. Each has a report (renders inline), a print-ready **PDF**, and an interactive **demo** (opens in your browser; see `../demos/` for download links too).

1/4 · Configuration · PDF · ▶ demo — two standard models of one network. ONF **TAPI** ↔ IETF **TEAS/ACTN**, seeded with false cognates and opaque items. The founding setting: it shows the thin reference **substituting for cognition**, the benefit **capability-dependent**, verification catching the traps, and work scaling **linearly** with a reference against quadratically without. Its four sub-studies are folded into the report and preserved as method notes under `notes/studies/`.

2/4 · Intent · PDF · ▶ demo — refinement, negotiation, and a service that renegotiates itself. A customer's intent reconciled against an operator's catalogue by **refinement**, not equivalence; verification becomes a **satisfaction** check; a two-sided **negotiation** appears; the pragmatic operation enters as a portable **movable policy**; and the exchange **recurs across a service's life**. Grounded in the IRTF NMRG draft *draft-janz-nmrg-naas-agentic-negotiation*.

3/4 · Cross-domain · PDF · ▶ demo — reconciling with no public standard. Two **home-grown, private** models meet at one seam. The central result is a **mirror**: without the constructed reference the strong agent **under-commits** (defers at perfect precision) and the weak agent **mis-commits** (binds wrongly). A single descriptive field unlocks a capable agent; a bare shared pointer is worse than nothing. Building the shared ground is the work.

4/4 · Observability · PDF · ▶ demo — an alarm is not an anomaly. A legacy fault manager and an IETF **NMOP** agent (RFC 9940) reconcile two observability worlds, carrying the programme's deepest false cognate and a one-to-many decomposition. The **pragmatics carry the operative verdict** — but only for an agent able to carry them; while **correlation**, a structural pragmatic, is robust across the whole ladder.

The reconciling agents are one **model ladder** for the whole study: `gpt-5.6-sol` (**strong**), `gpt-5-mini` (**mid**), `gpt-5-nano` (**weak**) — "sol / mini / nano" — chosen so the ladder isolates model strength rather than provider or architecture.

How it works

The **lift** is the move from a *data model* — a schema and its records, meaningful only to whoever wrote it — to a *semantic model*: the same elements given grounded meaning, so each concept carries a lexicon (labels and synonyms), an ontology (its kind and relations), definitions and worked examples, and concrete instances as evidence, with an optional binding to a shared reference. Being self-describing, a lifted model is **portable** — any cognitive consumer can pick it up with no pre-agreed standard. Reconciliation runs over these lifted models, not over labels.

A **case** is two lifted semantic models plus a gold standard *derived from the models and validated*, so it cannot drift. A **reasoning stack** reconciles them — deterministic controls and a language-model agent run at each point on the **cognition spectrum** (both live, one inert, both inert). The **harness** scores the output against the gold and records quality — *precision*, *resolved fraction*, *surviving false cognates*, and the *residual* — and, for the agent, cognitive effort (reasoning tokens, latency). A resolved fraction below one is deferral, not error: the residual is the shortfall from full cognition's reach, and it grows as cognition recedes.

Repository layout (`study/`)

```
reports/                      the deliverables
    MASTER_REPORT.md              the synthesis — start here
    REPORT_1of4_configuration.md  setting 1 · configuration (TAPI ↔ TEAS)
    REPORT_2of4_intent.md         setting 2 · intent (refinement, negotiation, lifecycle)
    REPORT_3of4_cross_domain.md   setting 3 · cross-domain (standard-free)
    REPORT_4of4_observability.md  setting 4 · observability (alarm ≠ anomaly)
notes/
    studies/                  setting-1 method notes (lift baseline, reference anatomy,
                              instance disambiguation, verification modes)
    design/                   design notes of record, per setting
    archive/                  superseded working notes
    SETUP_OPENAI.md           how to supply the API key (kept out of the repo)
figures/                      all report figures (regenerated by pipeline/figures*.py)
src/reconcile/                the harness: models, reference, metrics, stacks, oracles
benchmark/                    case builders and cases/ (two lifted models, reference, traps, gold)
results/                      the exact per-run data behind every figure and table
pipeline/                     all runner, figure, and build scripts
tests/                        offline tests of the stacks and scoring (no API)
pyproject.toml
```

Quick start

Run from the `study/` directory. The controls, gold derivations, offline tests, figures, and PDF build need no model access (standard-library Python plus `matplotlib`/`markdown`/`wkhtmltopdf`). The language-model runs need `openai` and an API key — see notes/SETUP_OPENAI.md — and reach `api.openai.com`, so they run where that is available.

```
cd study

# deterministic, no API:
python pipeline/run.py --case config_big_hard --no-write     # controls only
python pipeline/scaling.py --max-n 12                         # the scaling result
python -m pytest tests/                                       # offline tests
python pipeline/figures_master.py                             # regenerate the master figures
python pipeline/build_pdfs.py                                 # rebuild every report PDF

# the language-model studies (need OpenAI; run where api.openai.com is reachable):
python pipeline/run.py --case config_big_hard --agent --trials 4 \
  --placement both_cognitive,one_inert,both_inert \
  --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/intent_study.py     --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/pragmatics_study.py --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/observability_study.py --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
```

Per-setting run commands are in each report's Reproducibility section, and the method notes under `notes/studies/` carry the sub-study commands.

Grounding and scope

The study operationalizes and empirically tests the claims of the proposal: the IRTF NMRG work on agentic network-as-a-service negotiation, the cross-domain provisioning demonstration, and the IETF NMOP anomaly/alarm model (RFC 9940). The claims are **existential and mechanistic** — *this is how ad hoc reconciliation works, and here it is working* — established on single, seeded cases built to exercise each mechanism and prove each trap, across the model ladder; they are not population estimates. Natural next steps are larger and more varied cases, more rungs on the ladder, and the end-to-end construct-then-bind protocol in the standard-free setting.